



GraniteWMS Install Wizard

Operator Manual

Version 0.5.1

For GraniteWMS V6.0 and V7.0 releases

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Overview

The GraniteWMS Install Wizard installs the GraniteWMS core stack on a Windows server: Web Desktop, Business API, Custodian API and Process App, together with the Granite database, the IIS app pools and HTTPS sites, a certificate and Windows Firewall rules. It replaces Granite.Scaffolding.exe.

It works from a GraniteWMS release exactly as it's shipped, as a folder or as the release .zip, for both V6.0 and V7.0. Everything the server needs (IIS features, the URL Rewrite module and the .NET 8 Hosting Bundle) is installed from the release itself, so the server doesn't need internet access.

Nothing on the server changes until you press **Start Install** on the last screen, and a **dry run** there checks every setting first and shows exactly what would be done.

What it covers

- A fresh install on a new server, with a new, clean Granite database.
- An install that points the apps at an existing Granite database, keeping its data.
- Reinstalling over an earlier install on the same server (the old IIS sites are replaced and the old files kept as a backup).

What it does not cover (yet)

- Granite Scheduler, Telemetry, label printing and the ERP integration services.
- SSRS report server settings.
- Upgrading an existing V6.0 database to V7.0. For V7.0, install into a new database.

ABOUT THIS VERSION

This manual describes version 0.5.1. The version is shown in the wizard's title bar; check it matches the build you're testing, and quote it when reporting an issue.

Before You Begin

Have the following ready before starting the wizard on the target server:

- **Administrator rights** on the server. The wizard asks for elevation when it starts.
- **The GraniteWMS release**, V6.0 or V7.0, as a folder or as the release .zip, copied to the server (or on a share it can read).
- **SQL Server 2016 SP1 or later**, reachable from the server, with **Mixed Mode** (SQL Server and Windows) authentication. The Granite apps log in with a SQL login.
- **A SQL account with sysadmin rights** for the install itself: either the Windows account you're running the wizard as, or a SQL login such as sa. It's used only during the install and never saved.
- **The address users and scanners will use** to reach the server: its IP address, or a DNS name that resolves on the warehouse network.
- **Four free TCP ports**. The defaults are 40099 (Web Desktop), 40081 (Business API), 40082 (Custodian API) and 40080 (Process App).
- **A password for the Granite app login** (default login name Granite). The wizard can generate one. Write it down: it's never saved.
- **A backup of the database**, if you'll be pointing the install at an existing Granite database.

Getting the Wizard

The wizard is a single, self-contained .exe. There's nothing to install and no .NET runtime needed on the server beforehand.

- Copy **GraniteInstallWizard-v0.5.1.exe** to the server. Putting it next to the release folder or zip saves a step.
- Double-click it and accept the Windows administrator prompt.
- Check the version in the title bar.

If a folder called Granite V6.0 sits next to the exe, one level up, or in Documents\Granite WMS, the wizard finds it and fills it in on Step 1. Any other release, including a .zip, is picked on Step 1.

Step-by-Step Walkthrough

The wizard has six steps. Use **Next** and **Back** to move between them; answers are kept when you go back. Each step checks its own fields before letting you continue.

Step 1: Release and Install Folder

Tells the wizard which GraniteWMS release to install and where to put it.

Fields on this screen

- **Granite release:** the release folder (**Folder...**) or the release zip (**Zip file...**). The status line underneath turns green when a complete release is found. A zip is extracted to C:\ProgramData\Granite Install Wizard\Releases and reused if you pick the same zip again, so the first time takes a few minutes and later times are instant. V7.0 releases, where each app is its own zip inside the release, are unpacked there as well; your own folder is never changed.
- **Install folder:** where the apps are installed, default C:\Program Files\GraniteWMS. Each app gets its own subfolder. If the folder doesn't exist, click **Create folder**, or the wizard offers to create it when you click Next.
- **Company name:** shown in Web Desktop and Process App.
- **Date format:** DD/MM/YYYY, MM/DD/YYYY, YYYY/MM/DD or YYYY-MM-DD. Web Desktop and the Business API spell date formats differently; the wizard writes the right spelling to each.
- **Load saved profile...:** fills every answer from an earlier install except passwords. Every install saves a profile (InstallProfile.json) to its install folder, so the next server can start from the same answers.

STATUS LINE MESSAGES

"Found a complete Granite release" means you can continue. If the release is missing a part the wizard needs (one of the four app folders, the database create script, or GraniteScaffold\Prerequisites), the status line names what's missing.

Step 2: Prerequisites

Shows what the server already has. Anything missing is installed during the install, from the release's own GraniteScaffold\Prerequisites folder.

Fields on this screen

- **The status table:** IIS features, the IIS URL Rewrite module and the ASP.NET Core 8 Hosting Bundle, each marked OK or Missing. **Check again** refreshes it.
- **Install missing IIS features, Install the IIS URL Rewrite module and Install the ASP.NET Core 8 Hosting Bundle:** leave these ticked. If something required is missing and its box is unticked, the wizard won't continue.
- **Also install the ASP.NET Core 6 Hosting Bundle:** optional; the core stack doesn't use it.

RESTARTS

Installing the Hosting Bundle restarts IIS so it picks up the new module. If Windows asks for a server restart while installing features, the install finishes and the summary says so; restart the server before go-live.

Step 3: SQL Server

Connects to SQL Server, chooses the database, and sets the SQL login the Granite apps will use.

Fields on this screen

- **SQL Server / instance name:** pick a detected instance or type one, for example .\SQLEXPRESS or SQL01\GRANITE. **Refresh** searches again. Instances with the SQL Browser service off aren't listed but can still be typed.
- **Connect as:** Windows Authentication (the account running the wizard) or a SQL Server login. This account needs sysadmin: it creates the database, the app login and the SQLCLR assembly.
- **Test Connection:** must pass before you can continue. Each result is shown on its own line: green when connected, orange for something to note, red for something that stops the install. Test again after changing anything above it.
- **Granite database:** choose **Create a new, clean database** (runs the release's create script) or **Use an existing Granite database** (keeps its data; the create script is not run). For an existing database, the name list is filled after Test Connection with the databases on the server that have Granite's tables.
- **Drop and recreate the database if it already exists:** new-database mode only. Every row in the old database is lost, and the wizard asks you to confirm.
- **SQL login the Granite apps use:** default Granite. It's created if it doesn't exist and made db_owner of the database. Enter the password twice, or click **Generate**; tick **Show** to see it.
- **If this login already exists with a different password, change its password:** for reinstalls where the old password isn't known. Anything else using the login will need the new password, and the wizard asks you to confirm.
- **Run the Hotfix database scripts:** runs the release's Hotfix database scripts, listed by name on the checkbox. On by default for a new database, off for an existing one. Disabled when the release has none (V7.0).
- **Apply the Hotfix app files:** copies the release's updated app files over the installed ones, keeping each app's settings files. The checkbox lists which apps have Hotfix files.

IF TEST CONNECTION SHOWS RED

"Not sysadmin": connect with an account that has sysadmin, or have one granted for the install.

"Windows Authentication mode only": in SSMS, open the server's Properties, then Security, choose SQL Server and Windows Authentication mode, and restart the SQL Server service.

"Database already exists" (new-database mode): choose Use an existing Granite database, pick another name, or tick Drop and recreate.

"Login already exists, but the password entered doesn't work": enter the login's current password, or tick the change-its-password option.

Step 4: Websites

Sets the address users will type, and the IIS site name and HTTPS port for each app.

Fields on this screen

- **Address users and scanners will type:** pick one of the server's addresses or type a DNS name. An IP address is the safest choice for scanners on Wi-Fi without a DNS entry for the server. This address goes into the apps' settings and must be on the certificate (Step 5).
- **The site table:** one row per app, with a tick box, the IIS site and app pool name, the HTTPS port and the resulting URL. The Business API is always installed because the other three depend on it.
- **Open these ports in Windows Firewall:** adds an inbound TCP rule per port. Leave ticked unless the firewall is managed elsewhere.

- **Replace existing IIS sites with these names:** for reinstalling. Any IIS site with one of these names is removed with its app pool, SSL binding and firewall rule, and recreated. Its old files are kept (the folder is renamed to .bak-<date>). The wizard asks you to confirm.

App	Default IIS site name	Default HTTPS port
Web Desktop	Granite WebDesktop	40099
Business API	Granite Business API	40081
Custodian API	Granite Custodian	40082
Process App	Granite Process App	40080

PORTS ALREADY IN USE

A port held by another site or program is caught before anything is changed, on Step 6. If the old Granite sites are the ones using the ports, tick Replace existing IIS sites; otherwise choose different ports.

Step 5: HTTPS Certificate

All four sites run over HTTPS. Browsers and scanners only trust the certificate if it lists the exact name or IP address they use to reach the server.

Fields on this screen

- **Create a new self-signed certificate** (default): enter a friendly name, the DNS names and the IP addresses, each list comma-separated. They're pre-filled from the server and Step 4. The certificate is valid for 5 years.
- **Trust it on this server:** adds the certificate to this server's Trusted Root store, so the server's own browser doesn't warn.
- **Use an existing certificate from Local Computer\Personal:** pick one from the list (only certificates with a private key that haven't expired are shown); the names on it are listed underneath. **Refresh** reloads the list.

SCANNERS AND CLIENT PCS

A public copy of a new certificate (.cer) is saved to the install folder's Certificates subfolder. Install it on scanners and client PCs so they trust the sites. If the Step 4 address isn't on the certificate, the wizard warns before continuing.

Step 6: Review and Install

Shows every answer in one place, then runs the install.

Fields on this screen

- **The review:** release, install folder, prerequisites, database, websites and certificate. Use Back to change anything.
- **Dry run:** runs every check and logs what would change, without changing anything. The button then reads **Start Dry Run**. Do a dry run first on every server.
- **Start Install:** asks for confirmation, then runs the stages listed in the next section. A progress list shows each stage, with the full log underneath.

- **Cancel:** stops after the current step. Stages that finished stay done; the log says where it stopped.
- **Save profile...:** saves the answers (without passwords) to a file for another server.
- **Open log:** opens the log file for the current run.

RECOMMENDED ROUTINE

1. Tick Dry run and click Start Dry Run.
2. Read the log: every problem found is listed at once, and nothing has changed.
3. Fix anything reported, untick Dry run and click Start Install.

What Start Install Does

The install runs in ten stages, shown in order on Step 6. It stops at the first stage that fails and says where it stopped. After fixing the cause, running it again is safe: the checks recognise what's already in place.

Stage	What happens
1. Pre-flight checks	Checks everything before changing anything: administrator rights, the release and its create script, SQL Server access and rights, the database and login, IIS site names and ports, and the certificate. All problems are listed together.
2. Prerequisites	Installs whatever is missing: IIS features, then URL Rewrite, then the .NET 8 Hosting Bundle, all from the release. Restarts IIS after the Hosting Bundle.
3. HTTPS certificate	Creates the self-signed certificate (and trusts it, if ticked) or checks the chosen existing one. Saves the public .cer to the install folder.
4. Database	Creates or checks the app login, runs the create script for a new database, maps the login into the database as db_owner, then runs the Hotfix database scripts if ticked.
5. Copy application files	Removes the IIS sites being replaced (if ticked), renames any existing app folder to .bak-<date>, copies each app from the release, then applies the Hotfix app files if ticked.
6. Configure applications	Writes each app's appsettings.json: connection strings, the addresses the apps use to reach each other, allowed origins, company name and date format.
7. IIS app pools and sites	Creates an app pool (No Managed Code, always running) and an HTTPS site per app, binds the certificate, grants the app pool rights to its folder, then starts everything.
8. Firewall rules	Adds an inbound rule per port, if ticked.
9. Verify installation	Checks the app login can read the database, and that every site answers over HTTPS. Sites get about a minute to start.
10. Summary	Writes InstallSummary.txt and InstallProfile.json to the install folder.

Every run, including dry runs, writes a log to C:\ProgramData\Granite Install Wizard\Logs. Passwords are masked in the log.

After the Install Completes

The final message on Step 6 is one of:

- **GraniteWMS is installed.** Everything passed.
- **Installed, but some checks failed.** The install finished but a verification check didn't pass; the log shows which. Often a site that needed longer to start: browse to it and check.
- **Stopped with an error.** The stage that failed is named; fix the cause and run it again.

Then

- Open **InstallSummary.txt** in the install folder. It lists the URL of each site, the database and login, and the certificate.
- Sign in to Web Desktop at its URL (by default https://<address>:40099/).
- Install the certificate (.cer from the install folder's Certificates subfolder) on scanners and client PCs, then open Process App on a scanner (https://<address>:40080/).
- Restart the server if the summary says Windows asked for one.
- Keep **InstallProfile.json** for the next server; load it on Step 1.

Reinstalling and V7.0 Releases

Reinstalling over an earlier install

- Step 1: the same install folder as before. The existing app folders are renamed to .bak-<date>, not deleted.
- Step 3: choose **Use an existing Granite database** and pick it. Enter the app login's current password; if it isn't known, tick the change-its-password option.
- Step 4: keep the same site names and ports, and tick **Replace existing IIS sites**.
- Step 6: dry run first. The log lists each site and app pool that will be replaced.

V7.0 releases

- Pick **Granite V7.0.zip** (or the unzipped folder) on Step 1. V7.0 packs each app as a zip inside the release; the wizard unpacks only the parts the core stack needs, into C:\ProgramData\Granite Install Wizard\Releases.
- Install into a **new database**. The wizard does not upgrade a V6.0 database to V7.0.
- V7.0 ships no Hotfix database scripts the wizard runs, so that option is disabled on Step 3. The Hotfix app files (Business API and Process App) are applied as usual.
- To keep a V6.0 install running alongside V7.0 on the same server, give the V7.0 sites different names and ports on Step 4 (for example 41099, 41081, 41082 and 41080).

Troubleshooting Quick Reference

If you see...	Do this
"The zip contains an unsafe path and was not extracted"	The zip has an entry that would be written outside the extraction folder, so it's refused on purpose. Check the zip came from the official release. (Version 0.5.0 also showed this, wrongly, for Granite V6.0.zip; fixed in 0.5.1.)
Step 1 status line: "The release folder is missing: ..."	The folder isn't a complete release. Pick the top folder of the release (the one containing GraniteBusinessAPI, GraniteDatabase and so on), or the release zip.
Step 2: a required item is Missing and Next is refused	Leave its install option ticked. It's installed from the release during Step 6.
Test Connection: "is not sysadmin"	Connect as an account with sysadmin rights (for example sa, or a Windows admin account that's sysadmin on the instance).
Test Connection: "Windows Authentication mode only"	In SSMS: server Properties, Security, SQL Server and Windows Authentication mode. Restart the SQL Server service, then test again.
"Database ... already exists" on a new-database install	Choose Use an existing Granite database, pick another name, or tick Drop and recreate (this deletes the old data).
"Database ... doesn't look like a Granite database"	The database has no Granite tables. Check the name, or create a new database instead.
"Login ... already exists, but the password entered doesn't work"	Enter the login's current password, or tick the change-its-password option on Step 3.
Pre-flight: "An IIS site called ... already exists"	Tick Replace existing IIS sites on Step 4 to reinstall over it, or choose another site name.
Pre-flight: "Port ... is already used by the IIS site ..." or "by another program"	Choose another port on Step 4, or stop what's using it.
Verify: a site returns HTTP 500 or 503	Open the site in a browser on the server for the detailed error. Check the app's appsettings.json connection string and that the .NET 8 Hosting Bundle is installed, then recycle its app pool in IIS Manager.
Browsers or scanners show a certificate warning	The address being used isn't on the certificate, or the device doesn't trust it. Install the .cer from the install folder on the device, or reinstall with the address added on Step 5.
The summary says Windows asked for a restart	Restart the server before go-live.
Anything else	Click Open log on Step 6 (or look in C:\ProgramData\Granite Install Wizard\Logs) and send the log with the wizard version from the title bar.